

PROJECT PROGRESS REPORT 2

Automated Classification of Sports-Related Injuries via Medical Imaging

Reporting Period: Week 3–4 | Date: April 19, 2026 | Phase: Model Training & Evaluation

Task	Status	Notes
Stratified train / val / test split (70/15/15)	COMPLETE	6,204 train · 1,329 val · 1,330 test
PyTorch Dataset & DataLoader pipelines	COMPLETE	Separate loaders for baseline and ResNet50
Data augmentation (train only)	COMPLETE	Rotation $\pm 15^\circ$, H-flip, brightness/contrast jitter
Baseline CNN training (5 epochs)	COMPLETE	lr = 1e-3, Adam optimiser
ResNet50 transfer-learning fine-tuning (5 epochs)	COMPLETE	lr = 1e-4, ImageNet IMAGENET1K_V1 weights
Evaluation: accuracy, confusion matrix, F1, AUC	COMPLETE	Both models benchmarked on held-out test set
Prototype recommendation function	COMPLETE	Maps predicted class to clinical guidance text
Hyperparameter tuning / extended training	PLANNED	Scheduled for Progress 3
Grad-CAM / saliency visualisations	PLANNED	Explainability milestone in Progress 3

1. Overview

This report documents progress for Weeks 3 and 4 of the sports injury classification project. Building directly on the data acquisition and preprocessing foundations established in Report 1, this phase focused on (i) constructing reproducible stratified data splits and PyTorch DataLoader pipelines, (ii) training a custom baseline convolutional neural network (CNN), and (iii) fine-tuning a pretrained ResNet50 backbone via transfer learning. All core milestones for this phase have been completed on schedule.

2. Dataset & Stratified Splits

2.1 Dataset Continuity

The same **Hemg/bone-fracture-detection** dataset (Hugging Face) used in Progress 1 was retained to ensure comparability across milestones. The full training split contains 8,863 radiograph images across two balanced classes (fractured / not fractured), with an imbalance ratio of approximately 1.02 as confirmed by the EDA in Report 1.

2.2 Stratified Three-Way Split

A stratified split was implemented per class to preserve the ~50/50 balance in every partition. The resulting split sizes are shown below:

Split	Total Images	Fractured	Not Fractured	Ratio
Train	6,204	3,136	3,068	70 %

Validation	1,329	672	657	15 %
Test	1,330	672	658	15 %

3. DataLoader Pipelines & Augmentation

Two parallel transform chains were constructed — one for the baseline CNN (simple [0, 1] normalisation) and one for ResNet50 (ImageNet mean/std normalisation per the proposal). Train-time augmentation was applied exclusively to the training loader.

Pipeline	Resize	Augmentation (train only)	Normalisation
Baseline CNN	224 × 224	RandomRotation(15°), RandomHorizontalFlip, ColorJitter(brightness=0.15, contrast=0.15)	ToTensor() → [0, 1]
ResNet50	224 × 224	Same as above	ImageNet mean=(0.485, 0.456, 0.406) std=(0.229, 0.224, 0.225)

Both pipelines use **batch size = 32**, no persistent workers (NUM_WORKERS = 0 for Colab/Windows compatibility), and pin_memory enabled on CUDA devices.

4. Model Architectures

4.1 Baseline CNN (Custom)

A compact four-block convolutional network was implemented as the baseline, aligned with the proposal's description of a 3–4 convolutional layer design:

- Block 1: Conv2d(3→32, 3×3) + ReLU + MaxPool2d(2)
- Block 2: Conv2d(32→64, 3×3) + ReLU + MaxPool2d(2)
- Block 3: Conv2d(64→128, 3×3) + ReLU + MaxPool2d(2)
- Block 4: Conv2d(128→256, 3×3) + ReLU + AdaptiveAvgPool2d(1)
- Classifier: Linear(256 → 2)

4.2 ResNet50 (Transfer Learning)

A pretrained ResNet50 backbone with **IMAGENET1K_V1** weights was loaded via torchvision. Only the final fully-connected layer was replaced (in_features → 2). All layers were kept trainable during fine-tuning with a reduced learning rate of 1e-4 to preserve low-level feature representations from ImageNet pre-training.

5. Training Procedure

Both models were trained with **CrossEntropyLoss** and the **Adam** optimiser. Validation accuracy was tracked every epoch; the best checkpoint was restored before final test evaluation. The table below summarises the training configuration:

Parameter	Baseline CNN	ResNet50
Epochs	5	5
Learning rate	1e-3	1e-4

Optimiser	Adam	Adam
Loss	CrossEntropyLoss	CrossEntropyLoss
Batch size	32	32
Device	CUDA (if available) / CPU	CUDA (if available) / CPU

Note: 5-epoch runs are intended as a first-pass benchmark. Extended training and hyperparameter sweeps are planned for Progress 3.

6. Results & Evaluation

6.1 Quantitative Comparison

Both models were evaluated on the held-out test set (1,330 images). The following metrics were computed: accuracy, ROC-AUC, and per-class precision / recall / F1. Recall for the *fractured* class is treated as the primary clinical proxy (minimising false negatives / missed fractures).

Model	Test Accuracy	ROC-AUC	Recall (Fractured)	Recall (Not Fractured)
Baseline CNN	~51 %	~0.50	Limited — near chance	Limited — near chance
ResNet50	~98.6 %	~0.999	Very high	Very high

6.2 Baseline CNN Analysis

The custom CNN achieved near-chance accuracy (~51 %) after 5 epochs. This is consistent with expectations for a shallow network trained from random initialisation on a relatively small dataset (6,204 training images). Training loss decreased steadily, but validation loss remained high, indicating the model has not yet learned discriminative features within this training budget.

Key observations:

- Train loss decreased across epochs; val loss plateaued early — a sign of underfitting.
- Validation accuracy hovered near 50 %, consistent with predicting the majority class.
- Extended training (20+ epochs) and regularisation (dropout, weight decay) are expected to improve results.

6.3 ResNet50 Transfer Learning Analysis

The ResNet50 model achieved approximately 98.6 % test accuracy and a ROC-AUC of ~0.999 after just 5 fine-tuning epochs at $\text{lr} = 1\text{e-}4$. This demonstrates the significant advantage of leveraging ImageNet pretrained weights for medical image classification tasks where labelled data is limited.

Key observations:

- Epoch 1 val accuracy already reached 98.57 % — pretrained features transferred effectively.
- Near-perfect AUC indicates strong separation between fractured and non-fractured classes.
- Both recall values for fractured and not-fractured classes were high, minimising both false negatives (missed fractures) and false positives.
- Confusion matrix showed very few misclassifications on the 1,330-image test set.

6.4 ROC Curves

ROC curves were plotted for both models using the softmax probability of the *fractured* class (class index 0) as the injury score. The ResNet50 curve nearly reaches the top-left corner, confirming its near-perfect

discriminative ability. The baseline CNN's curve closely follows the diagonal (random classifier), reflecting the model's insufficient training at this stage.

7. Prototype Recommendation Function

As a proof-of-concept, a post-prediction recommendation function was implemented that maps the model's output class to high-level, non-clinical guidance text. This prototypes the downstream application envisioned in the proposal — where a trained model assists sports medicine staff with triage decisions:

- **Predicted fracture:** Flags urgent follow-up — immobilisation and specialist/surgical evaluation per local protocols.
- **Predicted no fracture:** Suggests clinical reassessment and graded return-to-play considerations, with a disclaimer that model output is not a substitute for a radiologist.

This function is intentionally minimal at this stage. Integration with a UI dashboard and confidence-threshold filtering will be explored in the final milestone.

8. Development Environment

Component	Details
Framework	PyTorch 2.x + torchvision
Python version	3.12 (Google Colab)
Hardware	NVIDIA GPU (Colab) / CPU fallback
Key libraries	NumPy, Pandas, Matplotlib, Scikit-learn, Pillow, HuggingFace datasets
Dataset source	Hemg/bone-fracture-detection (Hugging Face Hub)
Version control	Git (private repository)
Random seed	42 (fixed across PyTorch, NumPy, Python random)

9. Next Steps (Week 5–6)

- Extend training to 20+ epochs and implement learning rate scheduling (CosineAnnealingLR or ReduceLROnPlateau).
- Explore MobileNetV2 as a lightweight alternative to ResNet50 (proposal fallback model).
- Implement Grad-CAM saliency maps to visualise model attention regions and validate clinical plausibility.
- Investigate the MURA body-part metadata to introduce anatomy-site stratification (elbow, shoulder, wrist, knee).
- Apply class-weighted loss or focal loss if anatomy-site splits introduce class imbalance.
- Begin model deployment prototyping (Flask/Gradio inference endpoint).
- Document all findings, ablation comparisons, and visualisations in Progress Report 3.